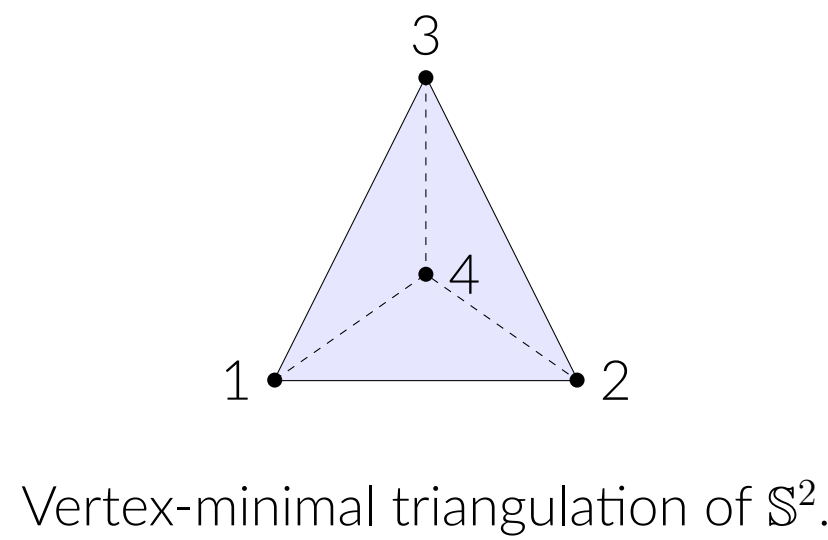


Triangulations

An (abstract) simplicial complex Δ is a collection of subsets of $[n]$ such that

- $\{i\} \in \Delta$ for all $i \in [n]$.
- if $S \subseteq T$ and $T \in \Delta$, then $S \in \Delta$.

Geometric realization $||\Delta||$:



Vertex-minimal triangulation of $\mathbb{S}^2$.

A simplicial complex is **centrally symmetric** if its vertex set has an involution $\iota : V \rightarrow V$ that induces a free action on all faces of Δ .

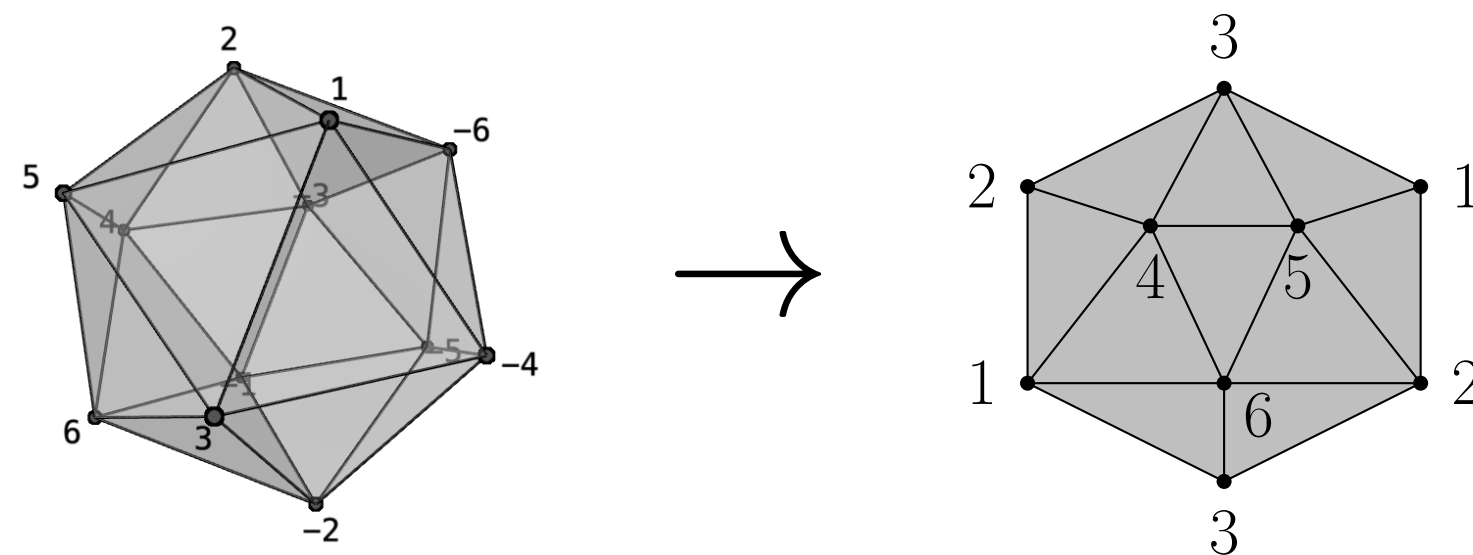
Polytopes

A **polytope** is the convex hull of finitely many points in $\mathbb{R}^d$.

It is **simplicial** if every facet is a simplex.

It is **centrally symmetric (cs)** when $x \in P$ if and only if $-x \in P$.

Many triangulations of $\mathbb{R}P^d$ arise by identifying antipodal vertices of a simplicial polytope.



Vertex-minimal triangulation of $\mathbb{R}P^2$.

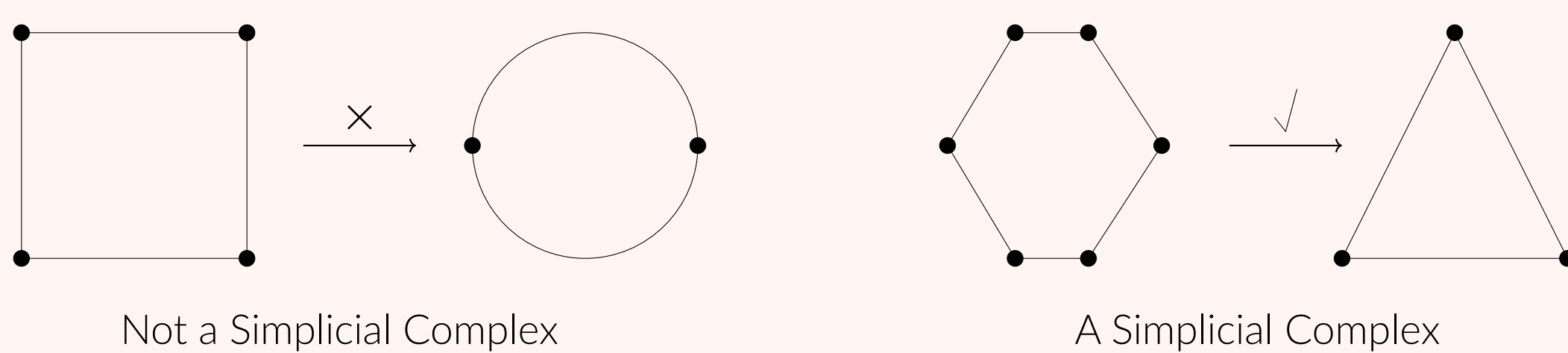
Goal: Construct a vertex-minimal triangulation of $\mathbb{R}P^5$ from a cs 6-polytope.

Requirements when Taking a Quotient

To obtain a valid triangulation of $\mathbb{R}P^d$ from a centrally symmetric triangulation Δ of $\mathbb{S}^d$, one can look at the graph of Δ . Namely, for every pair $v, -v$, we require that no vertex w is adjacent to both v and $-v$. That is, for every vertex v and antipode $-v$, we have

$$N(v) \cap N(-v) = \emptyset \quad (1)$$

This condition prevents the creation of multi-edges in the quotient.



Methodology

Key observation for a cs polytope P :

If $\langle u, v \rangle > 0$ for all edges $\{u, v\}$ of P , then $N(v) \cap N(-v) = \emptyset$ for all $v \in V(P)$.

1. Maximize the minimum inner product among any two vertices that share an edge. We run black-box optimization with AlphaEvolve with this objective to get a valid point set.
2. Exploit an " L^1 sparsity trick" borrowed from compressed sensing, to get a point set on $\mathbb{S}^5$ with "nice coordinates".
3. Verify the properties of this point set in SageMath.

The 6-Polytope $\mathcal{P}_{6,48}$

Coordinates of $\mathcal{P}_{6,48}$

For $\alpha = 3$, $\beta = 4$ and $\gamma = 5$, take the convex hull of following points in $\mathbb{R}^6$:

S = {1, 4, 5} $\pm(\gamma, 0, 0, \alpha, \beta, 0)$ $\pm(\alpha, 0, 0, -\gamma, \beta, 0)$ $\pm(-\gamma, 0, 0, -\alpha, \beta, 0)$ $\pm(-\alpha, 0, 0, \gamma, \beta, 0)$	S = {1, 2, 3} $\pm(\beta, \alpha, \gamma, 0, 0, 0)$ $\pm(\beta, -\gamma, \alpha, 0, 0, 0)$ $\pm(\beta, -\alpha, -\gamma, 0, 0, 0)$ $\pm(\beta, \gamma, -\alpha, 0, 0, 0)$	S = {2, 5, 6} $\pm(0, \beta, 0, 0, \alpha, \gamma)$ $\pm(0, \beta, 0, 0, -\gamma, \alpha)$ $\pm(0, \beta, 0, 0, -\alpha, -\gamma)$ $\pm(0, \beta, 0, 0, \gamma, -\alpha)$
S = {1, 4, 6} $\pm(\alpha, 0, 0, \gamma, 0, \beta)$ $\pm(-\gamma, 0, 0, \alpha, 0, \beta)$ $\pm(-\alpha, 0, 0, -\gamma, 0, \beta)$ $\pm(\gamma, 0, 0, -\alpha, 0, \beta)$	S = {2, 3, 4} $\pm(0, \gamma, \alpha, \beta, 0, 0)$ $\pm(0, \alpha, -\gamma, \beta, 0, 0)$ $\pm(0, -\gamma, -\alpha, \beta, 0, 0)$ $\pm(0, -\alpha, \gamma, \beta, 0, 0)$	S = {3, 5, 6} $\pm(0, 0, \beta, 0, \gamma, \alpha)$ $\pm(0, 0, \beta, 0, \alpha, -\gamma)$ $\pm(0, 0, \beta, 0, -\gamma, -\alpha)$ $\pm(0, 0, \beta, 0, -\alpha, \gamma)$

Vertices of $\mathcal{P}_{6,48}$ grouped by their supports.

$\mathcal{P}_{6,48}$ is centrally symmetric and lies on the 5-sphere of radius $\sqrt{50} = 5\sqrt{2}$.

Symmetries of $\mathcal{P}_{6,48}$

This polytope is highly symmetric. $\text{Aut}(\mathcal{P}_{6,48})$ has order 192 and is generated by:

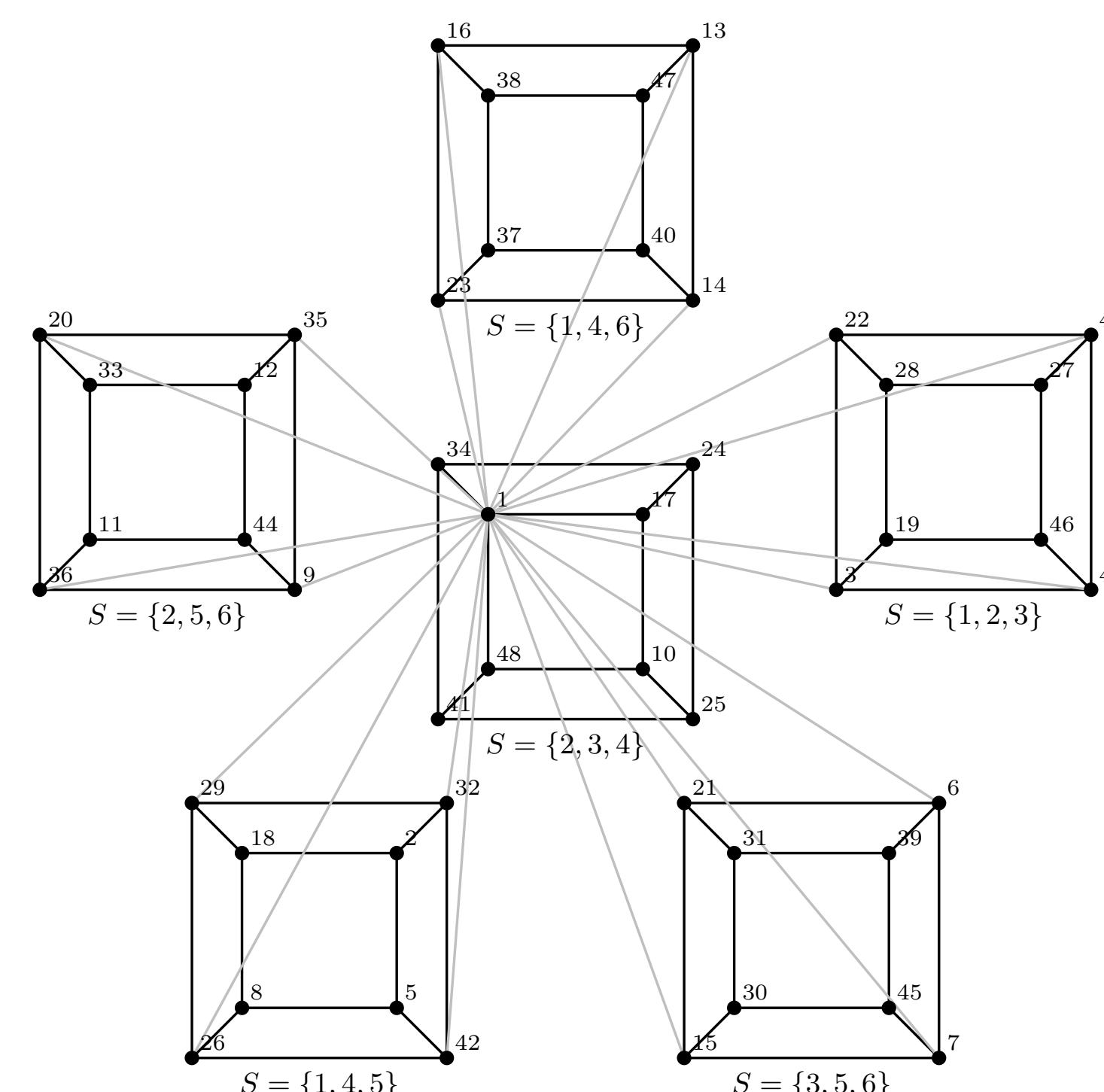
$$\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Generators of the Automorphism Group of $\mathcal{P}_{6,48}$.

Moreover, $\text{Aut}(\mathcal{P}_{6,48})$ acts transitively on $\mathcal{P}_{6,48}$.

Graph of $\mathcal{P}_{6,48}$

- Vertices of G : Vertices of $\mathcal{P}_{6,48}$
- Edges of G : two vertices u, v form an edge if and only if $\langle u, v \rangle > 0$.
- Consequently, G is 23-regular and vertex-transitive.



The union of induced subgraphs on V_S for every $S \in \mathcal{S}$ and the neighborhood of P_1 .

Prior Bounds for $\mathbb{R}P^d$

Small Dimensions

$d = 2$: The icosahedron provides the unique vertex-minimal triangulation on 6 vertices.

$d = 3$: Walkup [9] provides a 4-polytope whose quotient yields an 11-vertex triangulation and proves that it is minimal.

$d = 4$: Lutz [6] found the facet list of a 16-vertex triangulation with the BISTELLAR computer program [4]; Balagopalan [3] gave three ways to view this triangulation.

$d = 5$: Lutz [7] found the 676 facets of a 24-vertex triangulation with BISTELLAR. Lutz's example is not known to come from a polytope, and is distinct from our triangulation.

General Bounds

Upper Bound: Recently, Adiprasito–Avvakumov–Karasev [1] provided a subexponential construction with approximately $(2\sqrt{d})^{\sqrt{d}+1}$ vertices.

Lower Bound: For $d \geq 3$, any triangulation of $\mathbb{R}P^d$ requires at least $\binom{d+2}{2} + 1$ vertices and comes from Arnoux–Marin [2] using cup-length in cohomology.

Future Directions

Conjecture: The polytope $P_{6,48}$ yields a vertex-minimal triangulation of $\mathbb{R}P^5$.

- Every vertex w of $\mathcal{P}_{6,48}$ is adjacent to exactly one of $\{v, -v\}$ for any antipodal pair $v, -v$. Therefore, Condition (1) is tight for every antipodal pair of vertices.
- Neither our methods, nor the BISTELLAR approach of Frank Lutz [7], could find a triangulation with fewer than 24 vertices.

Question: Our methods improve the prior 53-vertex construction of Venturello–Zheng [8] for $\mathbb{R}P^6$ to 45 vertices. Is there a triangulation of $\mathbb{R}P^6$ with fewer vertices?

- Arnoux–Marin [2] gives a lower bound of 29. Therefore, there is likely room for further improvement.

Question: Our methods and prior constructions indicate that the Arnoux–Marin bound may not be tight. Can one find new methods to improve the lower bound?

- The best known construction of Adiprasito–Avvakumov–Karasev [1] is polytopal. Can one increase the lower bound for triangulations arising from quotients of polytopes?

Question: For balanced triangulations, the best known construction requires $\frac{3^{d+1}-1}{2}$ vertices while the lower bound also remains at that of Arnoux–Marin [2]. How can one improve either of these bounds?

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